

# Problem Statement: ClinsightAI

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## Objective

Build an AI-driven analytics system that helps healthcare business owners automatically extract structured operational insights and improvement recommendations from hospital review datasets.

The system should move beyond basic sentiment analysis and instead quantify drivers of ratings, detect recurring operational issues, and generate an actionable business improvement roadmap.

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# AI-Powered Healthcare Review Intelligence System

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## Problem Overview

Multi-location healthcare groups rely heavily on patient reviews to understand performance. However, review data is:

- Unstructured and noisy
- Difficult to quantify operational impact
- Hard to translate into action

Healthcare operators struggle to:

- Identify which operational themes most affect ratings
- Distinguish between systemic vs isolated issues
- Quantify the financial or rating impact of recurring problems
- Prioritize improvements that drive retention and profitability

Review datasets contain high-signal operational intelligence — but only if analyzed deeply and systematically.

# Challenge for Participants

Build an AI system that:

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## 1 Detects and Structures Operational Themes

Input:

- Hospital review dataset (provided Kaggle dataset)
- Star ratings
- Optional metadata (if available)

Output:

- Key recurring themes (e.g., wait time, staff behavior, billing, cleanliness, communication)
  - Clustered or labeled topic groups
  - Theme frequency distribution
  - Severity indicators
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## 2 Quantifies Impact on Ratings

Your system should:

- Measure correlation between themes and rating scores
- Identify which issues drive 1-star vs 5-star outcomes
- Estimate impact magnitude (regression, classification importance, etc.)
- Detect high-risk operational areas

Output:

- Ranked themes by impact on ratings
  - Quantified score contribution
  - Confidence score / robustness metrics
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## 3 Detects Recurring & Systemic Issues

- Identify repeated complaints across reviews
- Detect trend patterns (if time dimension used)
- Distinguish isolated outliers from systemic problems

Optional (advanced):

- Risk scoring system
  - Escalation scoring
  - Financial impact simulation
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#### **4 Generates Business Action Roadmap**

Your system must convert analytics into:

- Structured improvement recommendations
- Prioritized action plan
- “Quick wins” vs “High-effort improvements”
- Suggested KPIs to track

This is not just data analysis — this is decision intelligence.

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#### **5 Produces Structured, Explainable Output**

Your output should clearly explain:

- Why each theme was identified
  - Why it impacts ratings
  - How confident the model is
  - What action the clinic owner should take
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## Example JSON Output

```
{
  "clinic_summary": {
    "overall_rating_mean": 3.2,
    "primary_risk_themes": ["Long wait times", "Front desk behavior"]
  },
  "theme_analysis": [
    {
      "theme": "Wait Time",
      "frequency_percentage": 34,
      "rating_impact": -1.1,
      "severity_score": 0.82,
      "evidence_samples": [
        "Waited over 2 hours...",
        "Appointment delayed repeatedly..."
      ]
    }
  ],
  "improvement_roadmap": [
    {
      "priority": 1,
      "recommendation": "Optimize appointment scheduling workflow",
      "expected_rating_lift": "+0.6",
      "confidence": 0.78
    }
  ]
}
```

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## Dataset

Participants may use the Kaggle Hospital Reviews Dataset:

<https://www.kaggle.com/datasets/junaaid6731/hospital-reviews-dataset>

- ✓ Public dataset
- ✓ No PHI involved
- ✓ No scraping required

Teams may enhance with additional public healthcare review datasets if desired.

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## Expected Deliverables

- Working prototype (dashboard / notebook / app)
  - Clear explanation of methodology
  - Structured insight report
  - 5–7 minute demo presentation
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## Judging Focus

- Analytical depth
  - Quality & clarity of insights
  - Business actionability
  - Technical rigor
  - Visualization quality
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## Optional Advanced Features

- Embedding-based clustering (SBERT, OpenAI embeddings, etc.)
- Topic modeling (LDA / BERTopic)
- Rating prediction model
- Feature importance modeling
- Causal inference framework
- Profitability simulation engine
- Risk scoring dashboard
- Auto-generated executive summary

Kaggle link - <https://www.kaggle.com/t/7195ad2f88b04098a03747effdb68c61>